

BMN 616: ADVANCED OPTIMIZATION TECHNIQUES FOR MANAGERS

Spring 2026

Instructor:	Manu K. Gupta	Days:	Mon, Wed, Fri
Email:	manu.gupta@ms.iitr.ac.in	Time:	11:00 - 12:00 PM

Resources:

1. **Study material:** [Click Here](#)
2. **Virtual classes:** No Virtual Class.
3. **Teaching Assistant:** Abhinav Kumar, Email: abhinav_k@ms.iitr.ac.in,
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Course description: This course will be focused on optimization and will cover several aspects such as linear program, integer program, network optimization and large scale optimization. Google OR tools.

Target Audience: This course is primarily designed for MBA 2nd year students in operations or IT. This course is relevant in industry for roles such as Supply Chain Analyst, Operations Manager, Business Analysts, Data scientist, Business Consultant, Operations Research Analyst, Senior Analysts, Quantitative Researcher, IT manager and several other positions in management.

Prerequisites: Basic understanding of optimization. Students should be willing to learn coding in Python.

Goal: Upon completing this class, you will have a better understanding of several optimization models. You will also be able model and develop end solution for the problem in hand. Although you may not emerge as an expert, you will have gained the background to discuss and intelligently propose ways of applying optimization-based decision support system.

Office Hours: After class, or by appointment, or post your questions in the moodle.

Main References: The following are the major references for this course.

- Google OR tools. Link: <https://developers.google.com/optimization>
- Bertsimas, Dimitris, and Robert Michael Freund. *Data, models, and decisions: The fundamentals of management science*. South-Western Pub. 2004.

Grading Policy:

Open-laptop Quiz:	10%
Attendance:	10%
Demo Projects (Gurobi, 6 mins, Team of 2):	10%
Case based Project (Gurobi, 3 mins, Team of 1):	5%
Project Submission (Real, 9 mins, Team of 3):	15%
Final Exam:	45%
Class participation (basis quiz):	5%

Academic Honesty: Lack of knowledge of the academic honesty policy is not a reasonable explanation for a violation.

Project Topic: Students may choose one of the following two options for their group project:

Option A: 1. Find a problem from your surrounding/experience, etc. that can be modelled as an optimization problem, and solved in PYOMO.
2. The problem should be real, with real data and real problem constraints.

Option B: 1. Find a problem that is already documented in the literature. In this case, you are required to comprehend the problem and the associated mathematical (optimization) model, and reproduce the same in PYOMO. For this, you may look for problems from the following journals (which your library subscribes to):

- Approved case studies from INFORMS Transactions on Education.
- INFORMS Journal on Applied Analytics
- Operations Research
- European Journal of Operational Research
- Computers & Operations Research
- INFORMS Journal on Computing
- Transportation Science
- EURO Journal on Transportation and Logistics
- Transportation Research Part E: Logistics and Transportation Review

Option C: Use the real-data from the competitions that have been held in the past.

- <https://roadef.org/challenge/2026/en/>
- <https://coral.ise.lehigh.edu/mopta2026/competition/>
- <https://co-at-work.zib.de/#challenge>
- <https://registry.opendata.aws/amazon-last-mile-challenges/>
- <https://connect.informs.org/railway-applications/new-item3/problem-solving-competition6>
- <https://www.hackerearth.com/community/challenges/competitive/shell-hackathon/>
- <https://dimacs.rutgers.edu/programs/implementation-challenges/vehicle-routing>
- <https://connect.informs.org/tsl/tslresources/datachallenge>

The following is a tentative plan. Some of the problems to be taken up may change depending on the composition/interest of the class.

For Demo and Case based project – <https://www.gurobi.com/resources/case-studies>

Session	Problem/Case	Description/Objective	Readings
1 & 2	BlueSky Airline	To maximize revenue by controlling supply of limited seats across different routes	“Capacity-Based Revenue Management” in <i>Segmentation, Revenue Management & Pricing Analytics</i> by Bodea
3	BlueSky Airline (Cont.)	Introduction to PYOMO	PYOMO Documentation
4 & 5	Paper Recycling Problem	To determine the best combination of inputs and production processes for producing given outputs	Glasse, C.R. and Gupta, V.K., 1974. A linear programming analysis of paper recycling. <i>Management Science</i> , 21(4), pp.392–408.

Session	Problem/Case	Description/Objective	Readings
6 & 7	Workforce Scheduling	Non-linear versus Linear Formulations	Rothstein, M., 1973. Hospital man-power shift scheduling by mathematical programming. <i>Health Services Research</i> , 8(1), p.60.
8 & 9	Mars Incorporated Online Procurement	To determine the winning bids in a combinatorial auction	Hohner, G., Rich, J., Ng, E. 2003. Combinatorial and quantity-discount procurement auctions benefit Mars, Incorporated and its suppliers. <i>Interfaces</i> , 33(1), pp.23–35.
10	Knapsack Problem	Introduction to Branch & Bound Method for Integer Programs	Chapter 9, Winston
11 & 12	AmazingDeal.com	1. To determine the minimum set of fulfilment centers (FCs) to provide next day delivery to a given set of cities. 2. To determine the locations of a given number of FCs to cover the maximum demand.	1. Sankaran, J.K. and Raghavan, N.S., 1997. Locating and sizing plants for bottling propane in south India. <i>Interfaces</i> , 27(6), pp.1–15. 2. Aktaş, E., et al., 2013. Optimizing fire station locations for the Istanbul metropolitan municipality. <i>Interfaces</i> , 43(3), pp.240–255. 3. Van den Berg, P.L., et al., 2017. Increasing the Responsiveness of Firefighter Services by Relocating Base Stations in Amsterdam. <i>Interfaces</i> , 47(4), pp.352–361.
13 & 14	Freight Allocation with lane cost balancing	1. Sea Freight carrier selection and demand allocation. 2. Finding the appropriate value of BigM	Lim, A., Qin, H. and Xu, Z., 2012. The freight allocation problem with lane cost balancing constraint. <i>European Journal of Operational Research</i> , 217(1), pp.26–35.
15 & 16	Reynolds Construction	Project Scheduling & Crashing	Chapter 17, Stevenson
17	1. Don't Bother, I Can't Cope. 2. Executive Shirt Inc.	Assembly Line Balancing	Graves, S.C. and Lamar, B.W., 1983. An integer programming procedure for assembly system design problems. <i>Operations Research</i> , 31(3), pp.522–545.
18	Navi Mumbai Municipal Transport	1. Optimal EV Charging Installations. 2. Scheduling charging of EVs	Chapter 16, Stevenson
19	1. Choke Selection at Gas Wells. 2. Milk Collection Problem	1. Optimal Gas supply from gas wells to meet a given demand. 2. Optimal routing of truck to a given set of gas wells	Butler, M., Williams, H.P. and Yarrow, L.A., 1997. The two-period travelling salesman problem applied to milk collection in Ireland. <i>Computational Optimization and Applications</i> , 7(3), pp.291–306.
20	Product Line Selection	1. To determine the best subset of product configurations to develop. 2. Introduction to Bilevel Optimization	Jayaswal, S., and Sinha, A., 2023. Bilevel Optimization: Applications, Models and Solution Approaches
21	Student Project Presentations	—	—